Cross-Channel Dynamic Spatial–Spectral Fusion Transformer for Hyperspectral Image Classification

Zhao Qiu, Jie Xu, Jiangtao Peng, *Senior Member, IEEE*, and Weiwei Sun, *Senior Member, IEEE*

***Abstract*— Convolutional neural network (CNN) has achieved great success in hyperspectral image (HSI) classification. However, the local receptive field of CNNs leads to the drawback in extracting long-distance features. Transformer has excellent global modeling ability and shows good performance for HSI classification. The existing Transformer-based methods usually ignore the problem that the spatial information varies under different channels. To well describe the cross-channel dependencies, a cross-channel dynamic spatial–spectral fusion Transformer (CDSFT) is proposed in this article. In the proposed CDSFT, the multiscale and multichannel features are extracted and then cross-channel global features are extracted through transpose multihead self-attention (TMHSA). Next, a dynamic feature enhancement module and a spectral–spatial position attention (SSPA) module are designed to extract and enhance spectral–spatial joint features for classification. Experimental results on three well-known HSI datasets demonstrate the effectiveness of the proposed CDSFT method.**

***Index Terms*— Attention mechanism, convolutional neural network (CNN), hyperspectral image (HSI) classification, Transformer.**

I. Introduction

DIFFERENT from traditional optical images, hyperspectral images (HSIs) are composed of hundreds of consecutive spectral bands [1], [2]. They contain rich spectral and spatial information [3] and are widely used in many fields such as vegetation ecology [4], environmental ecology [5], geological survey [6], urban planning [7], target detection [8], and so on. HSI classification plays a key role in these fields. However, the large number of redundant spectral bands leads to the curse of dimensionality problem, which affects the classification performance of HSIs [9]. In the early research, to tackle the above-mentioned problem, a variety of dimensionality reduction methods have been proposed, such as principal component analysis (PCA) [10] and band selection [11]. These dimensionality reduction methods are usually combined with traditional machine-learning methods (e.g., Markov random field (MRF) [12], random forest (RF) [13], and support vector machines (SVMs) [14]) for classification. However, these traditional methods rely on handcrafted features and cannot extract data-dependent features.

To automatically learn features from hyperspectral data, deep-learning-based methods have been used for HSI classification and shown excellent performance [15]. Chen et al. [16] introduced a 3-D convolutional neural network (CNN) HSI classification method that uses 3-D convolution to learn both spatial and spectral features. Hamida et al. [17] introduced a new 3-D convolution model that extracts the joint spatial–spectral features and then extracts the spectral features by the 1-D convolution. Roy et al. [18] studied a hybrid CNN (HybridSN) for HSI classification, which first uses 3-D convolution to extract spectral–spatial joint features and then uses 2-D convolution to extract spatial features. There are also some methods to extract spatial features and spectral features separately. He et al. [19] considered a multiscale 3-D deep CNN (M3D-DCNN) to learn both 2-D multiscale spatial features and 1-D spectral features. Xu et al. [20] introduced a spectral–spatial unified network that separates spectral feature extraction, spatial feature extraction, and classifier training. However, as the number of network layers increases, it is prone to overfitting and leads to a decrease in classification accuracy in the CNN-based methods. To solve this problem, residual network structure has been incorporated into the CNN framework [21]. Zhong et al. [22] introduced a spectral–spatial residual network (SSRN) that can learn discriminative spectral–spatial features. Xu et al. [23] proposed a dual-channel residual network for the extraction of spectral and spatial features, respectively. Roy et al. [24] presented an adaptive spectral–spatial kernel residual network (A2S2KResNet) based on an attention mechanism to address the impact of convolution kernel size. Cui et al. [25] proposed a novel dual–triple attention network that captures the relevance between cross-dimension features.

In addition, in deep-learning-based methods, the network is generally a multilayer architecture [26]. How to exploit the dependencies between features under different layers or different channels is also a problem and worth exploring. Chen et al. [27] designed a cross-channel aggregation similarity network to calculate the similarity among features.

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Chen et al. [28] realized the information interaction of multi-order neighborhood nodes by fusing the network outputs from multiple layers. Yang et al. [29] presented a parallel, three-stage network model for hierarchical multiscale feature fusion. Wu et al. [30] proposed a cross-channel dense connection for extracting deeper spatial features. Although CNN-based methods have achieved good performance, there are still some problems. First, due to the fixed size of the local receptive field, only local features can be extracted. Second, the spectral features of HSIs have sequential attributes, and CNN-based methods have limitations in extracting spectral sequential features.

To address the aforementioned problems, many researchers focus on the Transformer model. The Transformer was first used in the field of natural language processing (NLP). It consists of an encoder and a decoder. With the emergence of vision Transformer (ViT) [31], it is proved that Transformers can also be applied to the field of computer vision. ViT divides the image into blocks and turns each block into a sequence. At the same time, a position embedding is generated and encoded together with the sequence by the encoder. Finally, the features learned with position embedding are used for classification. Many researchers have made improvements to the ViT. Liu et al. [32] designed a hierarchical ViT using shifted windows. Chen et al. [33] introduced a cross-attention multiscale ViT for image classification. Ren et al. [34] introduced a dynamic window ViT. To reduce the complexity of the Transformer, Ali et al. [35] designed a cross-covariance image Transformer.

Inspired by ViT and its variants, many researchers have applied them to HSI analysis. Hong et al. [36] introduced a Transformer for HSI classification and designed a SpectralFormer (SF) network. The network can learn local sequence information between adjacent spectra and transfer features from shallow to deep layers. To obtain spectral–spatial features and high-level semantic features, Sun et al. [37] combined the CNN and Transformer to propose a spectral–spatial feature tokenization Transformer (SSFTT), which adopts a Gaussian-weighted feature tokenizer to extract higher-level sequence features. Zhong et al. [38] proposed a spectral–spatial Transformer network (SSTN) that can solve the fixed-size convolution kernel problem through spatial attention and spectral association modules. To alleviate the problem of feature over-dispersion in the multihead self-attention (MHSA), Mei et al. [39] introduced a group-aware hierarchical Transformer (GAHT). Xie et al. [40] proposed a semantic and spatial–spectral feature fusion Transformer model to efficiently fuse semantic and spatial–spectral features.

Although the existing Transformer-based methods achieve good results, there are still some problems. First, these methods can solve the long-distance dependency problem and extract semantic features, but do not well exploit multiscale knowledge of HSIs. Second, they cannot handle cross-channel dependencies well. The spatial information of HSIs varies in different channels, and the convolution kernel size will have different effects on different channels. To well mine deeper spatial–spectral and cross-channel features, we propose a cross-channel dynamic spatial–spectral fusion Transformer (CDSFT) in the article. First, we extract cross-channel and multiscale local spatial features using convolutions. Then, local spatial features under different channels are processed by the transpose multihead self-attention (TMHSA). Next, dynamic feature enhancement and spectral–spatial position attention (SSPA) modules are designed to further enhance and highlight features. The proposed CDSFT takes advantage of the local feature extraction ability of CNNs and the global modeling ability of the Transformer to extract high-level semantic features. It can explore the dependencies between different channels. The main contributions of the article are summarized as follows.

1) A novel CDSFT is designed to mine, fuse, and highlight cross-channel and multiscale features. It uses multiscale convolution and cross-channel fusion to realize the interaction of information between channels.
2) A TMHSA is designed to reduce computational complexity by transposing query, key, and value matrices of self-attention. The transposed query, key, and value matrices will extract global features based on the relevance between features instead of sequences, which can improve classification accuracy.
3) A parallel, plug-and-play SSPA module is designed to highlight important spectral–spatial features. The spatial and spectral position information is exploited to generate attention weights to enhance the discriminative spectral–spatial features.

The remaining parts of the article are as follows. Section II introduces the proposed CDSFT method. Section III presents the experimental results. Section IV concludes the article.

II. Proposed Method

Fig. 1 shows the main architecture of the proposed method. First, the shallow spectral features are extracted by the spectral convolution module (SCM) and the generated feature maps are sliced into patch sequences by the patch embedding module. Then, the patch sequences are fed into the Transformer encoder and the SSPA module to extract and enhance spectral–spatial joint features. Finally, the spectral–spatial joint features are used to generate classification results.

A. Spectral Convolution Module

Due to the high dimensionality of HSIs, there are many redundant bands, so it is necessary to perform dimensionality reduction and extract effective spectral features. Considering that HSIs are essentially nonlinear, we use convolution operations for dimensionality reduction and simple feature extraction [41], [42]. Fig. 2 shows the implementation details of SCM, which extracts shallow spectral features through three convolution layers. Each convolution layer consists of a 1×1 2-D convolution operation, a batch normalization, and a ReLU activation function. The three convolution layers compress the channel dimension to 128, 96, and 64, respectively. The aim is to extract the local spectral features of HSIs.

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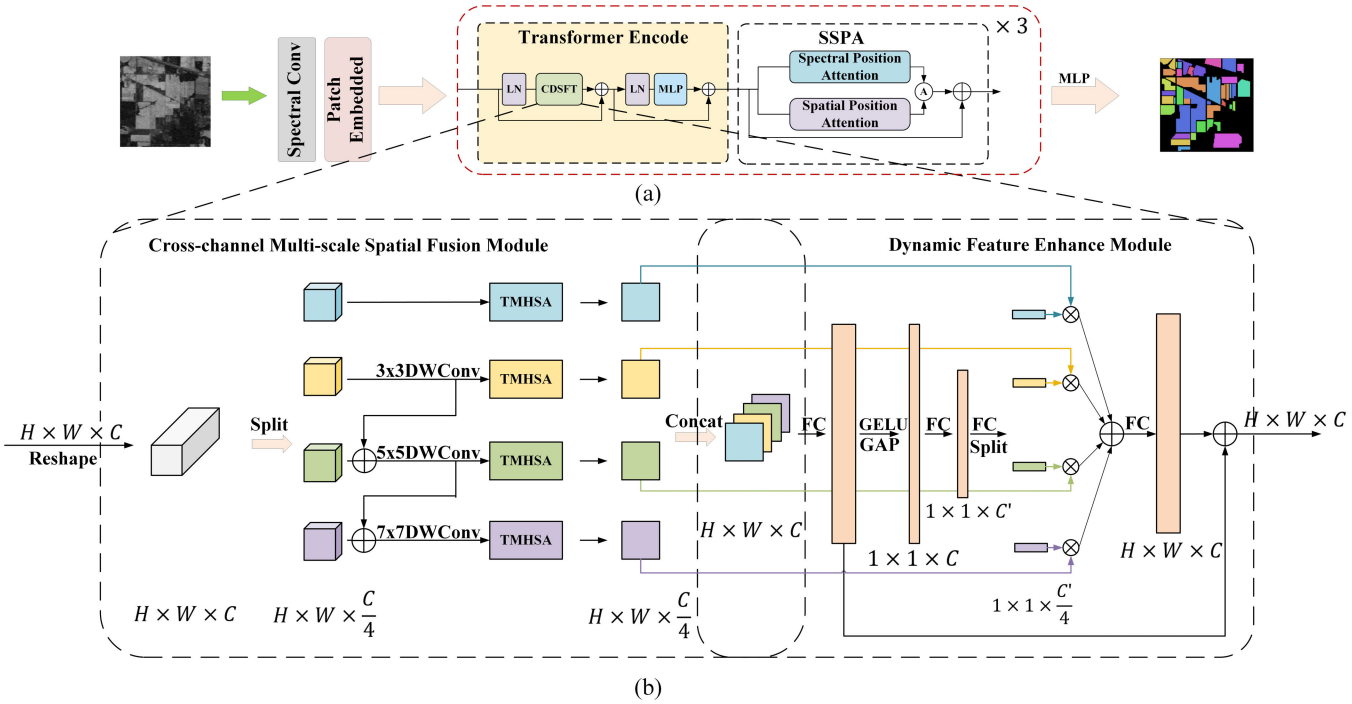


Fig. 1. (a) Framework of the proposed method, which mainly consists of three modules, that is, a spectral convolution and patch-embedded module, Transformer encode, and SSPA modules. (b) Framework of the CDSFT, consisting of a cross-channel multiscale spatial feature fusion module and a DFE module.

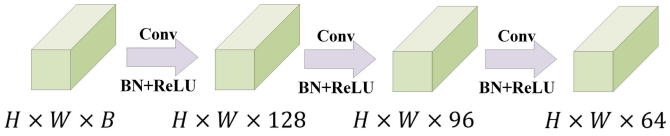


Fig. 2. Structure of SCM.

B. Cross-Channel Dynamic Spatial–Spectral Fusion Transformer

After the extraction of local spectral features in the SCM, the size of the HSI cube is $H \times W \times B$, where H and W are the height and width of the input HSI patch, respectively, and B is the size of the channel dimension after SCM. Then, the HSI cube $X \in \mathbb{R}^{H\times W\times B}$ is divided into patch sequence $x \in \mathbb{R}^{HW\times C}$ through the patch embedding module, where C is the length of the sequence. Next, x is fed into the CDSFT to further extract spectral–spatial global features. As shown in Fig. 1(b), the CDSFT consists of two modules, that is, cross-channel multiscale spatial fusion (CMSF) and dynamic feature enhance (DFE) modules.

1) Cross-Channel Multiscale Spatial Fusion: Consider that the spatial information is affected by input spatial patch size, and the spatial information varies in different spectral channels. We use the 2-D depth-wise convolution operation with different sizes under different channels to extract spatial local features. As shown in the left side of Fig. 1(b), the patch sequence x is reshaped to $I \in \mathbb{R}^{H\times W\times C}$. Then, I is sliced by the channel dimension, and $I_1, I_2, I_3, I_4 \in \mathbb{R}^{H\times W\times (C/4)}$ are obtained. Next, for I_1, its original features are kept. For I_2, a 3×3 depth-wise convolution is applied to extract local fine-grained spatial features. For I_3, we add the result of the convolution of I_2 to I_3 and use 5×5 depth-wise convolution to extract local medium-grained spatial features. For I_4, we add the result of the convolution of I_3 to I_4 and use 7×7 depth-wise convolution to extract local coarse-grained spatial features. Finally, the global features of the four branches are extracted by the TMHSA. We thus have the following output of CMSF:

$$\begin{aligned} I_i &= \text{Split}_i(I) \in \mathbb{R}^{H\times W\times \frac{C}{4}}, i = 1, 2, 3, 4 \\ y_1 &= \text{TMSHA}(I_1) \\ y_2 &= \text{TMSHA}(\text{Conv}_{3\times 3}(I_2)) \\ y_3 &= \text{TMSHA}(\text{Conv}_{5\times 5}(y_2 + I_3)) \\ y_4 &= \text{TMSHA}(\text{Conv}_{7\times 7}(y_3 + I_4)) \\ y_{\text{CMSF}} &= \text{CMSF}(I) = \text{Concat}(y_i). \end{aligned} \tag{1}$$

2) Dynamic Feature Enhance: With the CMSF module, we can get the output $y_{\text{CMSF}} \in \mathbb{R}^{H\times W\times C}$, which is used to generate dynamic weights for features of different channels. As shown in the right of Fig. 1(b), DFE is divided into two main steps: *fuse* and *select*. The former is to fuse the spatial global features under different channels, and the latter is to obtain the weights of the features under different channels. Then, the initial global features are multiplied by corresponding weights to dynamically perform global feature enhancement. The details of these two parts are as follows.

Fuse: It mainly contains two fully connected layers F_{fc}, two activation layers F_{act}, and a global pooling layer F_{gap}. The details are as follows:

$$\begin{aligned} y_F &= F_{fc1}(y_{\text{CMSF}}) \\ y_{\text{Fuse}} &= F_{\text{act}}(F_{fc2}(F_{\text{gap}}(F_{\text{act}}(y_F)))) \in \mathbb{R}^{1\times 1\times C'} \end{aligned} \tag{2}$$

where F_{act} is the GELU function and $C' = C/2$.

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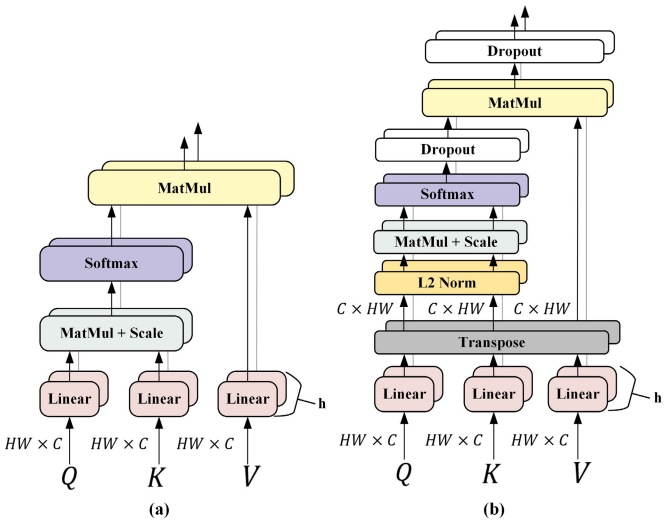


Fig. 3. Structure of (a) MHSA and (b) TMHSA.

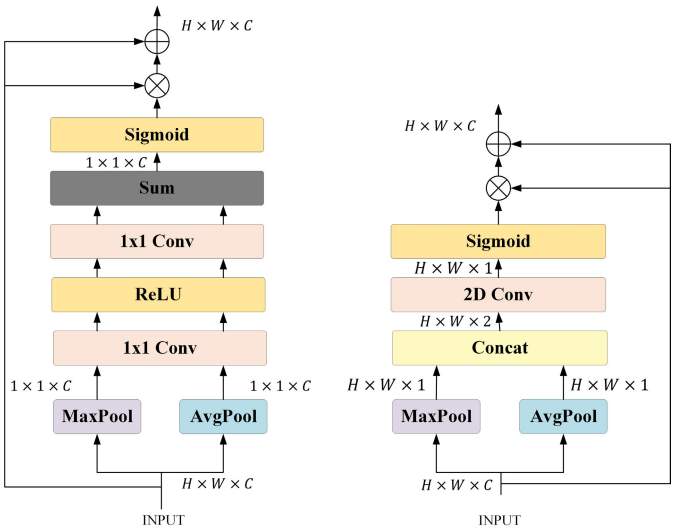


Fig. 4. Structure of (a) spectral position attention and (b) spatial position attention.

Select: It mainly contains two fully connected layers F_{fc} and a softmax layer Softmax. The details are as follows:

$$\alpha_i = \text{Softmax}\big(\text{Split}\big(F_{fc3}(y_{\text{Fuse}})\big)\big), \quad i = 1, 2, 3, 4$$
$$y_{\text{Select}} = F_{fc4}\left[\sum_{i}^{4}(\alpha_i \times y_i)\right] \in \mathbb{R}^{H\times W\times C} \tag{3}$$

where α_i is the weight for cross-channel feature y_i.

Finally, the output of DFE is as follows:

$$y_{\text{DFE}} = y_{\text{Select}} + y_F. \tag{4}$$

The output of the Transformer module is

$$y_t = \text{MLP}(\text{LN}(y_{\text{DFE}})). \tag{5}$$

3) Transpose Multihead Self-Attention: The main reason for the Transformer's great success is its MHSA mechanism. It can perform the computation of long-distance feature correlations and extract global features. For each head, three weight matrices W_q, W_k, and W_v are multiplied with the input sequences z to obtain three new sequences Q_i, K_i, and V_i (i is the ith head). Then $Q_i K_i^T$ is normalized and a softmax operation is performed to obtain the weights of each sequence. Finally, the weights are multiplied with V_i. The specific calculation processes are as follows:

$$Q_i = zW_q, \quad K_i = zW_k, \quad V_i = zW_v,$$
$$E_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d}}\right),$$
$$z_i = E_i V_i \tag{6}$$

where E_i and z_i are the weights of features and the output of the self-attention, respectively, and d is the dimension of K_i. Finally, the outputs of multiple heads are concatenated together as the final output.

Although the traditional MHSA operation can be used to extract global features, its complexity grows quadratically with the number of input patch sequences. As shown in Fig. 3(a), assuming that $Q, K, V \in \mathbb{R}^{HW\times C}$, the complexity of multiplying Q and K^T is $\mathcal{O}((HW)^2C)$. The resulting QK^T is further multiplied with V and the corresponding complexity is $\mathcal{O}((HW)^2C)$. Therefore, the overall complexity of MHSA is $\mathcal{O}(2(HW)^2C)$. Obviously, the complexity is squared with the height and width of the input patch. To reduce the computational cost, we transpose Q, K, and V and use the transpose MHSA (TMSHA) to extract features. As shown in Fig. 3(b), after transposing, the dimension of Q and K is $C \times HW$, and the overall complexity is $\mathcal{O}(2C^2HW)$, which is linear with respect to the size of the input spatial dimension. Although it has squared complexity for C, in our proposed method, the previous operation splits the channel dimension such that C is less than $H \times W$. For example, the patch sequence has a length of 128 and the size of C is 32 after splitting. To obtain sufficient spatial information, H and W are usually taken to be larger, such as $H = 9$ and $W = 9$. So, $\mathcal{O}(2HWC^2)$ will be smaller than $\mathcal{O}(2H^2W^2C)$. So, TMHSA reduces the complexity to some extent. In addition, by performing the transposition, TMSHA is based on features rather than sequences, which can generate attention feature maps with implicit knowledge about the global representation.

C. Spectral–Spatial Position Attention

To further enhance the spectral–spatial global features, we use spatial and spectral position information to calculate the weights of spatial and spectral features, respectively. As shown in Fig. 1, SSPA is divided into two branches, that is, spatial and spectral position attention branches.

1) Spectral Position Attention: As shown in Fig. 4(a), the maximum pooling and average pooling operations are performed on the input features to obtain spectral position information. Then, two 1×1 convolution operations are performed to obtain the spectral position features and the results are added together. Finally, the spectral position attention map is obtained by the sigmoid function. The details are as follows:

$$M_c = \sigma(\text{Conv}(MP(y_t)) + \text{Conv}(\text{AP}(y_t)))$$
$$y_c = M_c \times y_t + y_t \tag{7}$$

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where σ denotes the sigmoid function and M_c is the weight of the position of spectral information.

2) Spatial Position Attention: As shown in Fig. 4(b), the maximum pooling and average pooling operations are performed to compress the channel dimension to obtain spatial position information. Then, the results are concatenated into a feature map and a 2-D convolution operation is performed to extract spatial position features. The details are as follows:

$$M_s = \sigma\left(\text{Conv}\left(\left[\text{MP}(y_t); \text{AP}(y_t)\right]\right)\right)$$
$$y_s = M_s \times y_t + y_t \tag{8}$$

where M_s is the weight of the position of spatial information. Finally, the output of SSPA is

$$y_{\text{sspa}} = (y_c + y_s)/2 + y_t. \tag{9}$$

Algorithm 1 CDSFT Model

Input: Input an HSI data D and ground-truth Y; patch size $H \times W$; training sample rate r; the number of epochs e; the depth of model N;

Output: Output predicted labels.

```
Create HSI patch D' according to patch size H × W and divide the training set and testing set according to the rate of training samples r.
for i = 1 to e do
    Perform SCM to extract the shallow spectral features.
    Perform patch embedding and reshape to 3-D cube.
    for j = 1 to N do
        Perform CMSF to extract the local global spatial features.
        Perform DFE to enhance and fuse global features under different channels.
        Perform SSPA to highlight spectral–spatial features.
    end for
    Obtain classification result by MLP.
end for
Predict the corresponding labels.
```

Remark 1: Different from these algorithms [18], [19], [24], [36], [37], [38], [39], in Algorithm 1, we use convolution operations to extract local features under multichannels and perform information interaction between channels. Then we extract implicit global features of multichannels through improved self-attention operation.

D. Classification

After CDSFT and SSPA, the feature maps are flattened into 1-D vectors, and the classification result is obtained through the MLP module. Algorithm 1 demonstrates the overall algorithm flow.

III. Experiments

A. Datasets

1) Indian Pines: The Indian Pines (IP) dataset was captured by the AVIRIS sensor that covers agricultural areas in northwestern India. There are 220 bands in the original image, where 20 bands containing noise are removed and the remaining 200 bands are used. The spatial size is 145 × 145, and there are 16 mutually exclusive vegetation classes. The information is shown in Table I.

TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE IP DATASET

Class Number	Class Name	Training Number	Testing Number
1	Alfalfa	2	44
2	Corn-notill	71	1357
3	Corn-mintill	41	789
4	Corn	12	225
5	Grass-pasture	24	459
6	Grass-trees	37	693
7	Grass-pasture-mowed	1	27
8	Hay-windrowed	24	454
9	Oats	1	19
10	Soybean-notill	49	923
11	Soybean-mintill	123	2332
12	Soybean-clean	30	563
13	Wheat	10	195
14	Woods	63	1202
15	Buildings-Grass-Trees-Drives	19	367
16	Stone-Steet-Towers	5	88
	Total	512	9737

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE PU DATASET

Class Number	Class Name	Training Number	Testing Number
1	Asphalt	33	6598
2	Meadows	93	18556
3	Gravel	10	2089
4	Trees	15	3049
5	Painted metal sheets	7	1338
6	Bare Soil	25	5004
7	Bitumen	7	1323
8	Self-Blocking Bricks	18	3664
9	Shadows	5	942
	Total	213	42563

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE WHU-LK DATASET

Class Number	Class Name	Training Number	Testing Number
1	Corn	35	34476
2	Cotton	8	8366
3	Sesame	3	3028
4	Broad-leaf soybean	63	63149
5	Narrow-leaf soybean	4	4147
6	Rice	12	11842
7	Water	67	66989
8	Roads and houses	7	7117
9	Mixed weed	5	5224
	Total	204	204338

2) Pavia University: The Pavia University (PU) dataset was acquired by a reflective optical system imaging spectrometer (ROSIS) sensor. The spatial size is 610 × 340, and 12 bands containing noise are removed leaving 103 bands for the study. There are nine classes, and the information is shown in Table II.

3) WHU-LongKou: The WHU-LK dataset was acquired by the Headwall Nano-Hyperspec imaging sensor in Longkou Town, Hubei Province, China [43], [44]. The spatial size is 550 × 400 and the number of bands is 270. There are nine classes, and the information is shown in Table III.

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET

Class	M3D-DCNN	HybridSN	A2S2KResNet	SF	SSTN	SSFTT	GAHT	CDSFT
1	52.68 ± 17.42	93.81 ± 5.28	94.53 ± 6.64	45.91 ± 5.94	96.36 ± 7.27	97.77 ± 4.44	48.78 ± 18.58	**97.91 ± 2.58**
2	78.35 ± 5.61	97.28 ± 0.28	95.87 ± 1.45	81.62 ± 3.05	94.20 ± 8.53	97.97 ± 0.73	94.61 ± 1.92	**99.12 ± 0.19**
3	73.57 ± 10.66	95.18 ± 1.61	94.40 ± 2.19	78.24 ± 3.24	93.14 ± 5.27	95.05 ± 0.39	94.94 ± 1.62	**97.36 ± 1.39**
4	69.95 ± 8.65	98.81 ± 1.35	95.86 ± 2.90	79.92 ± 5.15	**99.81 ± 0.36**	99.34 ± 0.64	91.55 ± 2.57	99.72 ± 0.36
5	86.30 ± 4.82	95.74 ± 1.72	98.01 ± 1.41	82.13 ± 2.63	96.63 ± 2.46	97.31 ± 2.16	93.06 ± 3.19	**98.66 ± 0.43**
6	98.84 ± 0.64	97.45 ± 0.27	98.68 ± 1.27	89.51 ± 2.40	99.09 ± 0.32	98.23 ± 1.11	**99.70 ± 0.25**	99.36 ± 0.34
7	86.40 ± 10.31	**100 ± 0**	80.94 ± 20.46	25.70 ± 13.58	95.55 ± 5.44	84.45 ± 9.90	87.20 ± 14.18	95.99 ± 5.23
8	98.74 ± 1.71	97.81 ± 1.26	**99.77 ± 0.35**	92.48 ± 0.56	96.46 ± 2.76	95.28 ± 1.59	99.63 ± 0.43	94.86 ± 0.64
9	63.33 ± 18.79	96.31 ± 4.79	65.73 ± 20.92	51.15 ± 7.81	**100 ± 0**	92.29 ± 9.68	31.11 ± 11.44	97.94 ±2.51
10	73.85 ± 5.01	94.44 ± 0.33	94.69 ± 2.61	80.40 ± 1.57	94.73 ± 4.24	96.18 ± 0.70	94.47 ± 1.23	**96.62 ± 0.41**
11	86.96 ± 5.01	95.23 ± 1.25	96.89 ± 1.00	89.88 ± 1.21	94.15 ± 5.47	97.42 ± 0.68	94.12 ± 1.97	**98.27 ± 0.32**
12	71.65 ± 8.84	94.26 ± 2.29	96.46 ± 1.82	75.10 ± 2.18	95.37 ± 2.77	96.21 ± 48	94.31 ± 2.67	**98.09 ± 0.83**
13	98.81 ± 0.79	99.59 ± 0.58	97.54 ± 2.31	91.00 ± 3.25	98.99 ± 1.26	99.20 ± 1.34	99.35 ± 1.05	**99.59 ± 0.38**
14	97.09 ± 1.50	97.63 ± 2.27	97.39 ± 1.22	94.15 ± 0.93	97.72 ± 1.85	97.59 ± 0.43	**98.09 ± 1.56**	96.56 ± 0.85
15	79.71 ± 6.50	98.62 ± 1.90	95.27 ± 2.81	83.54 ± 3.00	97.50 ± 1.18	**98.98 ± 0.61**	91.24 ± 4.95	98.71 ± 0.94
16	84.52 ± 14.44	93.43 ± 2.94	90.36 ± 9.09	91.55 ± 5.24	98.59 ± 1.62	97.70 ± 1.56	94.52 ± 3.81	**99.29 ± 0.57**
OA	84.51 ± 2.20	96.21 ± 0.17	96.35 ± 0.66	85.52 ± 0.71	95.21 ± 2.56	97.15 ± 0.09	94.99 ± 0.45	**97.92 ± 0.16**
AA	81.30 ± 2.17	96.60 ± 0.48	93.27 ± 0.94	77.02 ± 1.65	96.77 ± 1.45	96.31 ± 0.62	87.92 ± 1.47	**98.00 ± 0.45**
κ	82.25 ± 2.56	95.68 ± 0.20	95.84 ± 0.75	83.51 ± 0.82	94.53 ± 2.92	96.74 ± 0.10	94.29 ± 0.50	**97.63 ± 0.18**

B. Experimental Setups

The proposed CDSFT method is compared with the following methods, that is, M3D-CNN [19], HybridSN [18], A2S2KResNet [24], SF [36], SSTN [37], SSFTT [38], and GAHT [39]. Among these methods, M3D-CNN, HybridSN, and A2S2KResNet are classical convolution-based methods, where SF, SSTN, SSFTT, and GAHT are recently proposed Transformer-based models. For a fair comparison, the parameter setting is the same as their original setting. We choose 5%, 0.5%, and 0.1% labeled samples per class to form training sets for IP, PU, and WHU-LK, respectively. We repeat the experiments ten times to get the mean and standard deviation as the final results.

All methods are implemented using PyTorch 1.10.1 and trained with an NVIDIA Quadro RTX 5000 GPU (16-GB memory). The batch size and the number of epochs are set to 128 and 200, respectively. The Adam optimizer is used and the learning rate is 0.001. The results of each model are evaluated in terms of three commonly used evaluation metrics, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ).

C. Comparison Experiments

Tables IV–VI show the mean OA, AA, and κ for ten runs on the IP, PU, and WHU-LK datasets, respectively. As the results demonstrate, our proposed method outperforms other methods. Specifically, on the IP dataset, our CDSFT obtains the OA value of 97.92%, which achieves an improvement between 0.77% for SSFTT and 13.41% for M3D-DCNN. The AA value is 98.00%, which also improves other methods by at least 1.23%. This demonstrates that the classification performance of CDSFT is relatively more balanced. And, the value of κ is 97.63%, which is also higher than other methods. In Classes 2, 3, 10, 11, and 12, CDSFT is significantly better than other methods. Classes 2 and 3 are subclasses of corn, and Classes 10, 11, and 12 are subclasses of soybeans. These classes have similar spectral characteristics so that they are difficult to distinguish. The proposed CDSFT can extract cross-channel spatial features, and the fusing of spatial and spectral features helps to discriminate spectrally similar classes.

To better demonstrate the experimental results, we show the classification map of all methods. As shown in Fig. 5, we zoomed in on a part of Class 11. Obviously, M3D-DCNN and SF have a high misclassification rate. HybridSN and A2S2KResNet misclassify Class 11 to Class 3. SSTN, SSFTT, GAHT, and CDSFT are Transformer-based methods, which misclassify Class 11 to adjacent classes. It is clear that our proposed method has a smaller misclassification rate.

On the PU dataset with 0.5% training samples, CDSFT also has more accurate classification results. In particular, CDSFT produces 98.26% of OA, 98.19% of AA, and 97.69% of κ. It gets an improvement of OA between 1.37% for A2S2KResNet and 13.36% for SF. For Class 3, the OA of CDSFT is 4.53% higher than the second most effective HybridSN method. This is because the distribution of Class 3 is relatively sparse and is susceptible to interference from surrounding pixels during classification.

We also show the classification map on the PU dataset. As shown in Fig. 6, we zoomed-in view a part of Class 3. It is clear that the map of CDSFT is closer to the ground truth. The other methods basically misclassify Class 3 into Class 8.

On the WHU-LK dataset, CDSFT achieves 98.50% of OA, 97.30% of AA, and 98.03% of κ. The standard deviation of OA, AA, and κ are 0.29%, 0.41%, and 0.39%, respectively, which are lower than other methods. It demonstrates that our CDSFT is more stable and effective. For Classes 3, 5, 8, and 9, which are relatively concentrated in distribution, our CDSFT can achieve higher accuracy with fewer samples.

As shown in Fig. 7, we zoomed-in view a part of the area close to Classes 4, 5, 8, and 9. This region has four classes and is susceptible to interference from other classes. Apparently, the proposed CDSFT has a much lower misclassification rate and is closer to the ground truth.

To further validate the effectiveness and robustness of the proposed method, we investigate the OA of different methods under different numbers of training samples. We construct

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TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PU DATASET

Class	M3D-DCNN	HybridSN	A2S2KResNet	SF	SSTN	SSFTT	GAHT	CDSFT
1	90.32 ± 3.12	90.79 ± 1.18	95.32 ± 1.35	82.90 ± 2.49	92.70 ± 3.05	92.05 ± 0.37	91.79 ± 2.22	**95.97 ± 1.04**
2	95.24 ± 1.58	96.61 ± 0.45	97.54 ± 0.52	88.64 ± 1.84	96.77 ± 0.41	**99.24 ± 0.37**	96.26 ± 1.40	99.17 ± 0.56
3	82.83 ± 4.90	94.39 ± 3.52	88.72 ± 4.94	58.47 ± 12.74	88.99 ± 4.55	82.45 ± 6.11	64.88 ± 11.38	**98.92 ± 0.92**
4	96.28 ± 2.97	97.14 ± 1.21	98.87 ± 0.42	93.26 ± 3.95	**99.76 ± 0.15**	92.77 ± 2.19	96.83 ± 0.72	97.68 ± 0.58
5	99.58 ± 0.41	94.83 ± 3.30	97.84 ± 2.11	98.45 ± 0.72	99.74 ± 0.22	97.93 ± 1.61	**99.88 ± 0.10**	99.03 ± 0.44
6	94.12 ± 1.56	98.67 ± 1.14	99.54 ± 0.48	92.15 ± 2.80	99.24 ± 0.24	95.78 ± 2.00	96.49 ± 2.22	**99.93 ± 0.05**
7	95.83 ± 2.09	85.85 ± 1.60	98.59 ± 0.81	71. 23 ± 2.22	98.56 ± 1.20	92.59 ± 2.98	93.02 ± 3.26	**98.91 ± 0.69**
8	82.48 ± 4.62	87.40 ± 4.09	95.18 ± 1.18	65.36 ± 3.06	89.39 ± 5.17	86.18 ± 1.58	84.52 ± 4.69	**95.27 ± 0.51**
9	99.07 ± 0.79	90.94 ± 4.50	**99.20 ± 0.86**	88.37 ± 15.19	98.06 ± 0.53	96.48 ± 1.67	95.44 ± 2.75	98.80 ± 0.33
OA	92.82 ± 1.14	94.43 ± 0.25	96.89 ± 0.30	84.90 ± 1.50	95.66 ± 0.95	94.97 ± 0.56	92.71 ± 0.58	**98.26 ± 0.36**
AA	92.86 ± 0.96	92.96 ± 0.61	96.76 ± 0.38	82.09 ± 2.69	95.91 ± 0.96	92.83 ± 1.22	91.01 ± 1.52	**98.19 ± 0.35**
κ	90.41 ± 1.53	92.56 ± 0.34	95.86 ± 0.40	79.50 ± 2.11	94.20 ± 1.28	93.34 ± 0.73	90.29 ± 0.76	**97.69 ± 0.48**

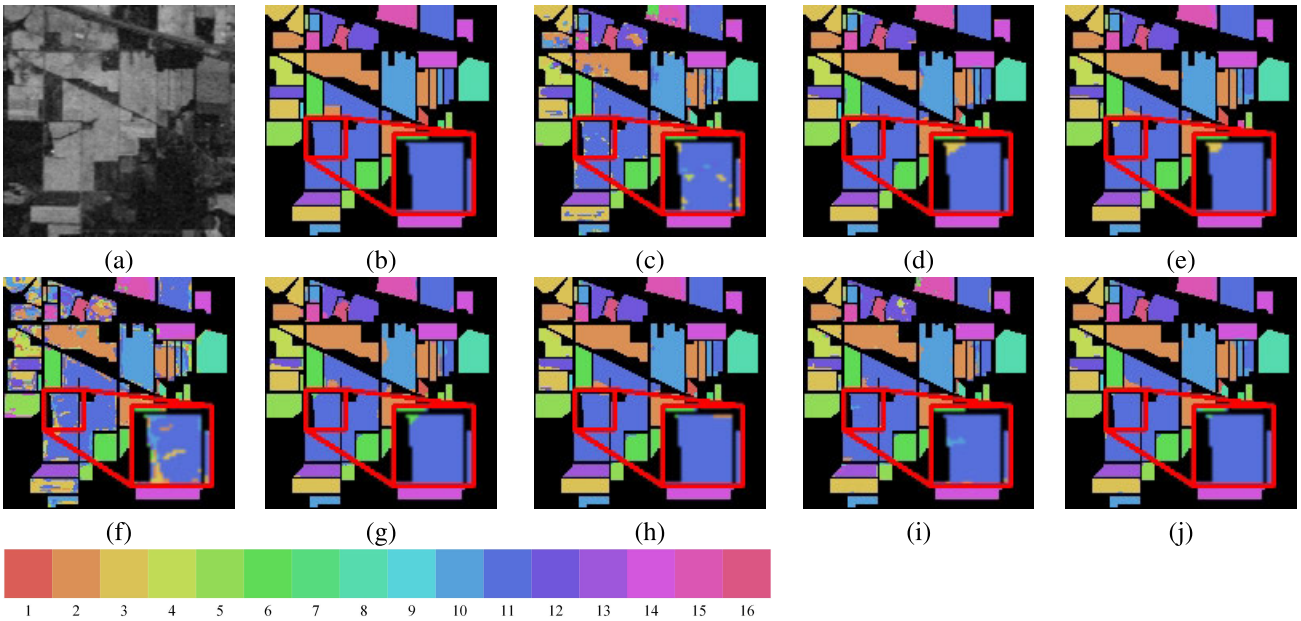


Fig. 5. Classification maps of the IP dataset. (a) False-color image. (b) Ground truth. (c) M3D-DCNN. (d) HybridSN. (e) A2S2KResNet. (f) SF. (g) SSTN. (h) SSFTT. (i) GAHT. (j) CDSFT.

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE WHU-LK DATASET

Class	M3D-DCNN	HybridSN	A2S2KResNet	SF	SSTN	SSFTT	GAHT	CDSFT
1	96.16 ± 0.44	96.30 ± 1.52	97.90 ± 1.12	96.15 ± 0.75	**98.86 ± 0.82**	98.54 ± 0.75	98.77 ± 0.64	98.45 ± 0.56
2	79.58 ± 5.74	**99.26 ± 0.58**	94.13 ± 7.09	70.20 ± 4.74	94.22 ± 6.22	94.15 ± 3.10	86.90 ± 8.06	98.86 ± 0.90
3	89.06 ± 7.81	94.28 ± 4.10	90.39 ± 4.15	80.15 ± 7.50	98.93 ± 0.73	94.23 ± 4.18	89.26 ± 4.15	**99.27 ± 0.59**
4	93.32 ± 0.23	**98.43 ± 0.57**	96.92 ± 1.00	94.13 ± 0.65	97.03 ± 1.68	97.03 ± 0.88	94.27 ± 1.53	98.20 ± 0.64
5	86.75 ± 8.16	91.97 ± 1.48	96.16 ± 2.53	58.37 ± 1.90	90.36 ± 9.48	91.07 ± 8.33	82.40 ± 4.46	**98.22 ± 1.22**
6	93.16 ± 3.07	97.99 ± 1.28	98.79 ± 0.82	86.30 ± 1.43	95.68 ± 2.14	97.78 ± 1.71	97.64 ± 0.95	**99.88 ± 0.06**
7	98.89 ± 0.24	99.21 ± 0.32	99.46 ± 0.22	98.92 ± 0.29	99.71 ± 0.30	99.67 ± 0.10	99.26 ± 0.43	**99.75 ± 0.08**
8	78.05 ± 3.42	78.47 ± 6.01	87.50 ± 5.77	74.03 ± 3.85	89.08 ± 3.07	81.32 ± 7.82	81.28 ± 6.42	**92.84 ± 3.24**
9	88.57 ± 3.65	88.78 ± 5.67	86.65 ± 7.97	51.72 ± 16.23	83.90 ± 7.84	88.95 ± 6.79	85.80 ± 3.84	**90.24 ± 2.31**
OA	94.28 ± 0.57	97.19 ± 0.64	97.18 ± 0.48	92.48 ± 0.66	97.26 ± 0.45	97.10 ± 0.57	95.54 ± 0.53	**98.50 ± 0.29**
AA	89.28 ± 1.92	93.85 ± 1.05	94.21 ± 0.88	78.88 ± 2.54	94.19 ± 1.64	93.64 ± 2.13	90.62 ± 0.74	**97.30 ± 0.41**
κ	92.44 ± 0.75	96.30 ± 0.85	96.28 ± 0.63	90.09 ± 0.85	96.38 ± 0.60	96.18 ± 0.75	94.11 ± 0.71	**98.03 ± 0.39**

the training set by selecting 1%, 2%, 3%, 4%, and 5% label samples per class for the IP dataset, and 0.1%, 0.2%, 0.3%, 0.4%, and 0.5% label samples per class for the PU and WHU-LK datasets, respectively. As shown in Figs. 8–10, the OA of CDSFT is higher than other methods for three datasets. In the case of a small number of samples, CDSFT still achieves good results. Specifically, for the IP dataset, CDSFT is significantly better than the second-best SSTN when the training sample ratios are 1% and 2%. For the PU dataset, CDSFT is consistently better than other methods. For the WHU-LK dataset, the advantages of CDSFT are particularly apparent when training sample ratios are 0.1% and 0.2%. The results of the three datasets demonstrate that the proposed method is more effective and robust.

D. Friedman Test

To prove the significant difference between our proposed method and other methods, we perform the Friedman Test [45] on OA, AA, and κ in Table VII. Taking the OA as

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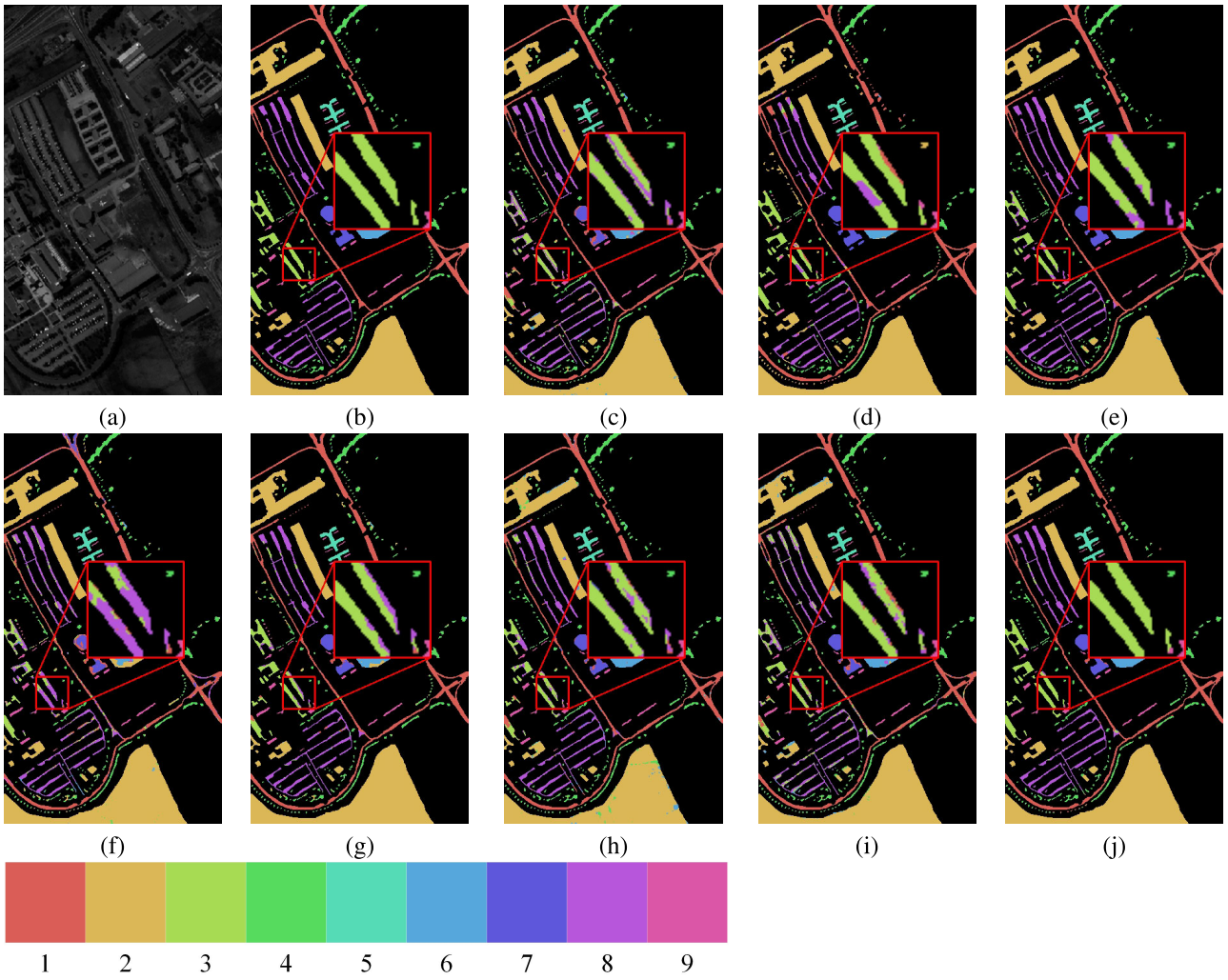


Fig. 6. Classification maps of the PU dataset. (a) False-color image. (b) Ground truth. (c) M3D-DCNN. (d) HybridSN. (e) A2S2KResNet. (f) SF. (g) SSTN. (h) SSFTT. (i) GAHT. (j) CDSFT.

TABLE VII
FRIEDMAN TEST OF OA, AA, AND κ

Metrics	Dataset	M3D-DCNN	HybridSN	A2S2KResNet	SF	SSTN	SSFTT	GAHT	CDSFT
OA	IP	8	4	3	7	5	2	6	1
	PU	6	5	2	8	3	4	7	1
	WHU-LK	7	3	4	8	2	5	6	1
	Average Rank	7	4	3	7.67	3.33	3.67	6.33	1
AA	IP	7	3	5	8	2	4	6	1
	PU	5	4	2	8	3	6	7	1
	WHU-LK	7	4	2	8	3	5	6	1
	Average Rank	6.33	3.67	3	8	2.67	4	6.33	1
κ	IP	8	4	3	7	5	2	6	1
	PU	6	5	2	8	3	4	7	1
	WHU-LK	7	3	4	8	2	5	6	1
	Average Rank	7	4	3	7.67	3.33	3.67	6.33	1

an example, we substitute the average rank value of OA in Tables IV–VI into (10) (n is 3 and m is 8) and get the observed value of τ_{χ^2} as 18.23. Then, we substitute the observed value of τ_{χ^2} into (11) and obtain the observed value of τ_F as 13.15. According to τ_F obeying $F_{(m-1,(m-1)(n-1))}(\alpha)$ distribution, and the significant level $\alpha = 0.05$. We query the table and give $F(7, 14)$ as 2.76. Since 13.15 is greater than 2.76, the H_0 hypothesis about OA is rejected. Similar results can be obtained for the AA and κ. Since H_0 is rejected, there is a significant difference between the proposed method and

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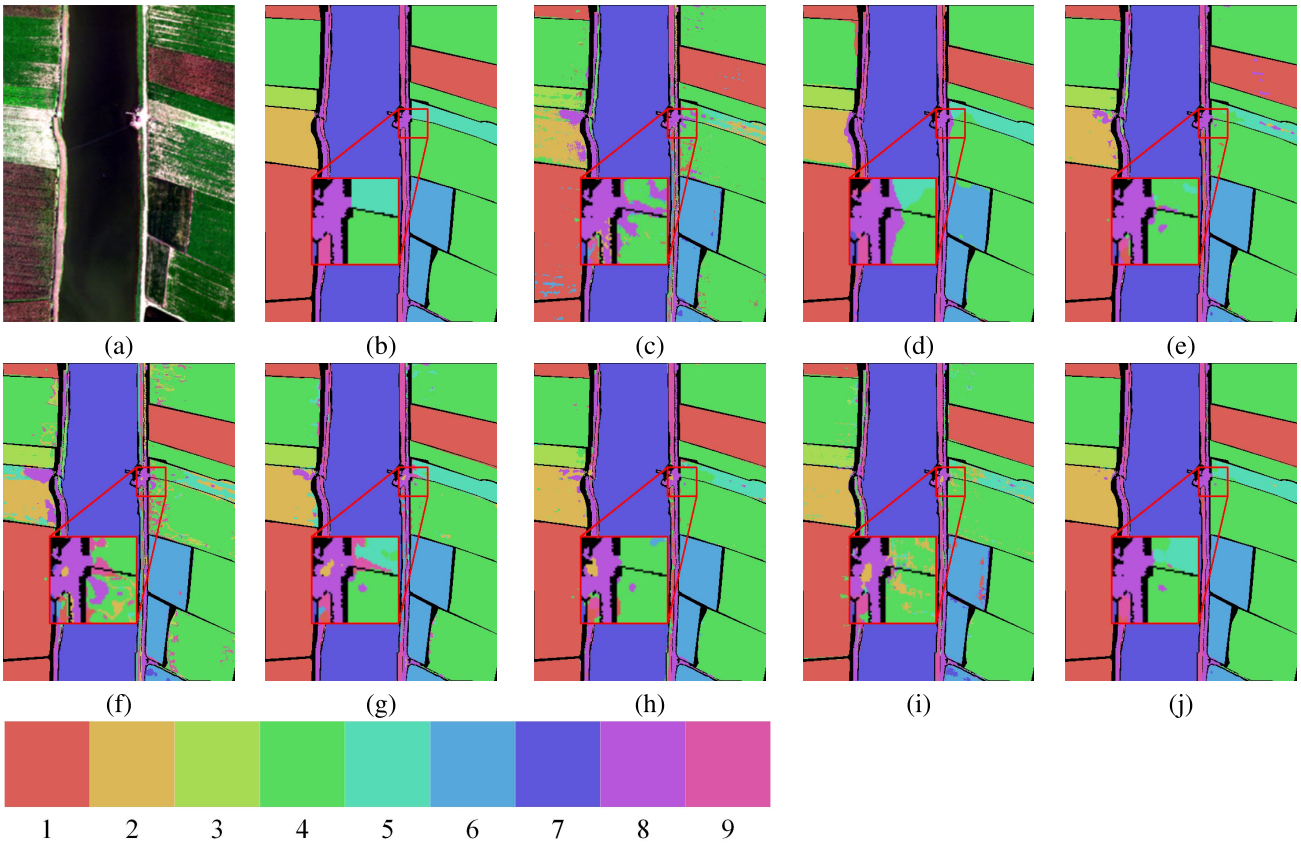


Fig. 7. Classification maps of the WHU-LK dataset. (a) False-color image. (b) Ground truth. (c) M3D-DCNN. (d) HybridSN. (e) A2S2KResNet. (f) SF. (g) SSTN. (h) SSFTT. (i) GAHT. (j) CDSFT.

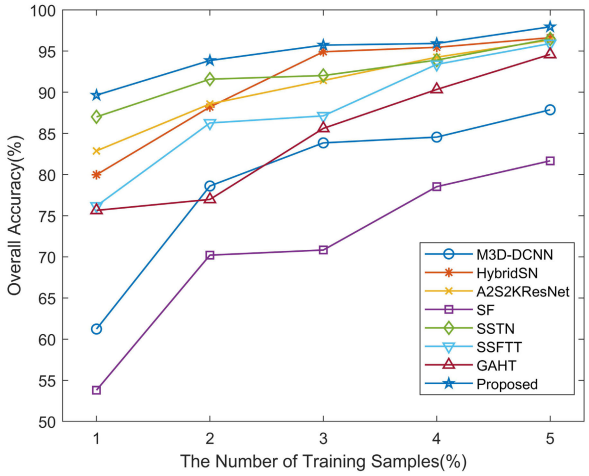


Fig. 8. Classification results versus different ratios of training samples on the IP dataset.

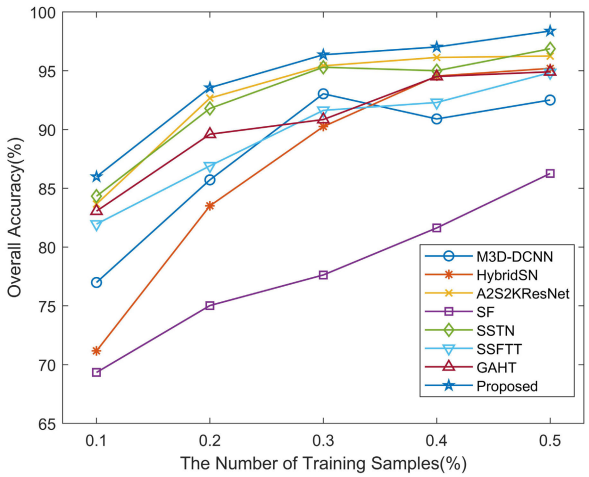


Fig. 9. Classification results versus different ratios of training samples on the PU dataset.

comparison methods. Our method has a rank value of 1, so it is the most effective

$$\tau_{\chi^2} = \frac{m-1}{m} \cdot \frac{12n}{m^2-1} \sum_{i=1}^{m} \left(R_i - \frac{m+1}{2} \right)^2 = \frac{12n}{m(m+1)} \left(\sum_{i=1}^{m} R_i^2 - \frac{m(m+1)^2}{4} \right) \tag{10}$$

$$\tau_F = \frac{(n-1)\tau_{\chi^2}}{n(m-1) - \tau_{\chi^2}}. \tag{11}$$

E. Ablation Experiments and Discussions

1) Ablation Analysis: The ablation experiment is performed to analyze the effect of different modules (i.e., CMSF, DFE, TMHSA, and SSPA) in the proposed CDSFT in Table VIII. When the CMSF is not selected, the feature maps under all channels are extracted using convolution kernels of the same size. When the DFE module is not used, the result of the CMSF module will be directly output. When the TMHSA module is not used, the original MHSA is used. From Table VIII, we can see that the OA value is the lowest when all

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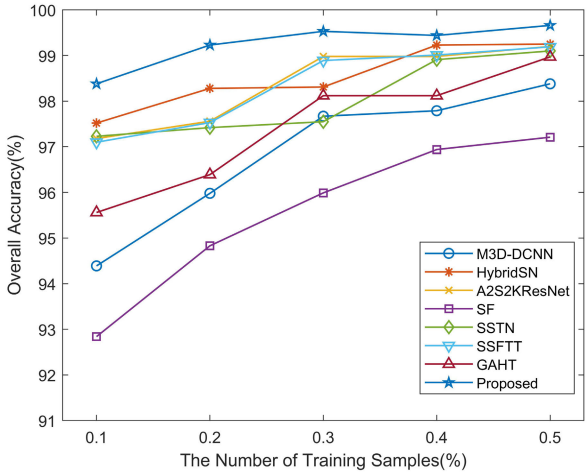


Fig. 10. Classification results versus different ratios of training samples on the WHU-LK dataset.

TABLE VIII
OA (%) UNDER DIFFERENT MODULES IN THE IP, PU, AND WHU-LK DATASETS

CMSF	DFE	TMHSA	SSPA	IP	PU	WHU-LK
×	×	×	×	95.54	95.31	95.69
✓	×	×	×	96.94	96.99	97.39
×	✓	×	×	96.28	96.05	96.19
✓	✓	×	×	97.55	97.36	98.11
✓	✓	✓	×	97.80	98.14	98.35
✓	✓	×	✓	97.76	97.57	98.29
✓	✓	✓	✓	**97.92**	**98.26**	**98.50**

TABLE IX
OA (%) AT DIFFERENT SCALES UNDER THE IP, PU, AND WHU-LK DATASETS

Dataset	IP	PU	WHU-LK
Single-scale	96.58	97.07	96.82
Multi-scale No-fusion	97.16	97.47	98.02
Multi-scale fusion	**97.92**	**98.26**	**98.50**

four modules are not selected. When CMSF and DFE modules are added, the OA on each dataset is increased by 2.01%, 2.05%, and 2.42%, respectively. This proves that multiscale cross-channel fusion of spatial information is more beneficial to feature modeling. Moreover, the OA has been improved by adding DFE or CMSF alone, which proves that DFE and CMSF play their role. After replacing the original MHSA with TMHSA, the OA values are also improved, indicating that the extraction of global features by self-attention between features is more effective than self-attention between sequences. Finally, with the addition of the SSPA module, the extracted features are further enhanced by SSPA. So, the classification performance is further improved.

2) Discussion on the CMSF: To verify the effectiveness of CMSF, we experiment with single-scale features, multiscale no-fusion, and multiscale fusion. For single-scale, we apply convolution operations of the same size 5×5 to all channels. For multiscale no-fusion, we remove the addition of features between channels. As shown in Table IX, the multiscale fusion

TABLE X
OA (%), TRAINING TIME (S), AND TESTING TIME (S) OF MHSA AND TMHSA

Dataset	Metrics	MSHA	TMSHA
IP	OA	97.76	**97.92**
	Training Time	91.26	**84.10**
	Testing Time	3.96	**3.84**
PU	OA	97.57	**98.26**
	Training Time	79.47	**59.93**
	Testing Time	29.28	**24.62**
WHU-LK	OA	98.29	**98.50**
	Training Time	88.31	**55.95**
	Testing Time	165.46	**127.35**

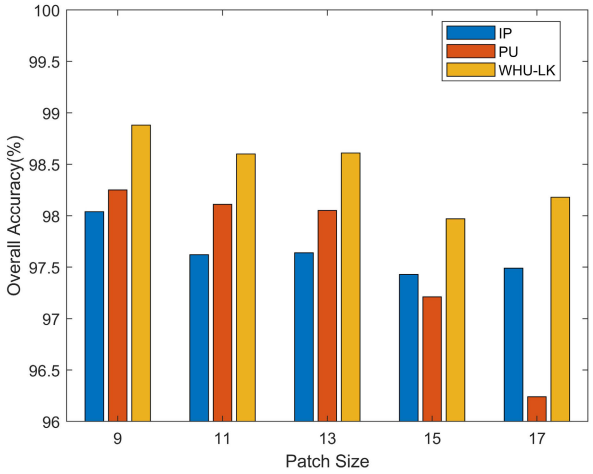


Fig. 11. OA (%) versus different patch sizes.

achieves the highest OA on three datasets, which verifies the validity of the CMSF.

3) Discussion on the TMHSA: To verify the validity of TMHSA, we conducted a comparison experiment between TMHSA and the original MHSA. The OA, training time, and testing time are compared and the corresponding results are shown in Table X. Compared with MHSA, TMHSA reduces the training time and improves the classification accuracy of all three datasets. In particular, for the WHU-LK dataset with 204542 samples, the training and testing time of TMHSA are 30 s faster than MHSA. The improvement of OA also demonstrates that global modeling of features instead of sequences is more effective for HSI classification.

4) Computational Cost Analysis: The training time, testing time, and the number of parameters of different methods on the three datasets are shown in Table XI. It can be seen that our CDSFT has higher computational efficiency than A2S2KResNet and SF and is slightly slower than SSTN, SSFTT, and GAHT. The high model complexity of CDSFT is mainly because of the repetition of the Transformer and SSPA modules three times in the network framework. To reduce the number of parameters, we use 2-D depth-wise convolution and TMHSA to improve computational efficiency. It is worth noting that although SSTN has the lowest number of parameters, the training time is not the lowest. This is because SSTN

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TABLE XI
COMPUTATIONAL COST ON THE IP, PU, AND WHU-LK DATASETS

		M3D-DCNN	HybridSN	A2S2KResNet	SF	SSTN	SSFTT	GAHT	CDSFT
IP	Training Time(s)	68.94	59.60	341.05	105.43	49.99	42.75	64.82	84.10
	Testing Time(s)	3.02	2.69	6.22	6.25	2.00	2.06	2.69	3.84
	Params(M)	0.59	3.54	0.37	0.35	0.021	0.93	1.43	1.12
	OA(%)	84.51	96.21	96.35	85.52	95.21	97.15	94.99	97.92
PU	Training Time(s)	22.50	20.57	65.42	31.13	25.81	19.31	26.51	59.93
	Testing Time(s)	7.85	6.84	15.42	14.85	6.34	5.73	8.59	24.62
	Params(M)	0.17	1.75	0.22	0.17	0.016	0.48	0.92	1.11
	OA(%)	92.82	94.43	96.89	84.90	95.66	94.97	92.71	98.26
WHU-LK	Training Time(s)	41.85	41.76	221.79	70.91	30.59	29.51	38.60	55.95
	Testing Time(s)	92.47	80.26	176.24	187.32	59.46	60.71	77.46	127.35
	Params(M)	0.46	4.83	0.48	0.55	0.024	1.25	1.51	1.13
	OA(%)	94.28	97.19	97.18	92.48	97.26	97.10	95.54	98.50

uses MHSA multiple times, resulting in a high computational cost. For SSFTT and GAHT, they need to fine-tune the hyperparameters of the model for different datasets, resulting in a large change in the number of model parameters. Although our proposed CDSFT needs more computational time, it does not need to fine-tune hyperparameters for different datasets and can achieve better classification results.

F. Effect of Patch Size

As the maximum convolution kernel size in the proposed CDSFT is 7×7, the spatial patch sizes of 9×9 to 17×17 are investigated. Fig. 11 shows the OA of CDSFT versus different spatial patch sizes on the IP, PU, and WHU-LK datasets. It is obvious that the best classification results are achieved at 9×9 for the IP, PU, WHU-LK datasets.

IV. CONCLUSION

In this article, we propose a CDSFT method for HSI classification. Different from other Transformer-based models, we consider the difference in spatial information between channels. So, we split the channel dimension and adopted different convolutional operations for different spectral channels to realize multiscale spatial feature extraction. We further analyze the correlation between the features under different channels and use a dynamic enhancement method to fuse and enhance the features under multichannels. In this way, the model can realize cross-channel interactions. In the proposed CDSFT, the TMHSA is used for global feature extraction, which is computationally efficient. In addition, SSPA is designed to further highlight important spectral–spatial features, which helps to learn discriminative features. Compared with some state-of-the-art deep-learning methods on the Indian Pines, Pavia University, and WHU-LongKou datasets, our CDSFT achieves better classification results.

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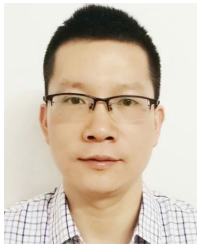


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